

Render ~ a mix of binder (cement) and fine aggregate (sand) with the addition of water and lime or a plasticiser to make the mix workable. Applied to walls as a decorative and/or waterproofing treatment.

Mix ratios ~ for general use, mix ratios are between 1:0.5:4–4.5 and 1:1:5–6 of cement, lime and sand. Equivalent using masonry cement and sand is 1:2.5–3.5 and 1:4–5. Unless a fine finish is required, coarse textured sharp sand is preferred for stability.

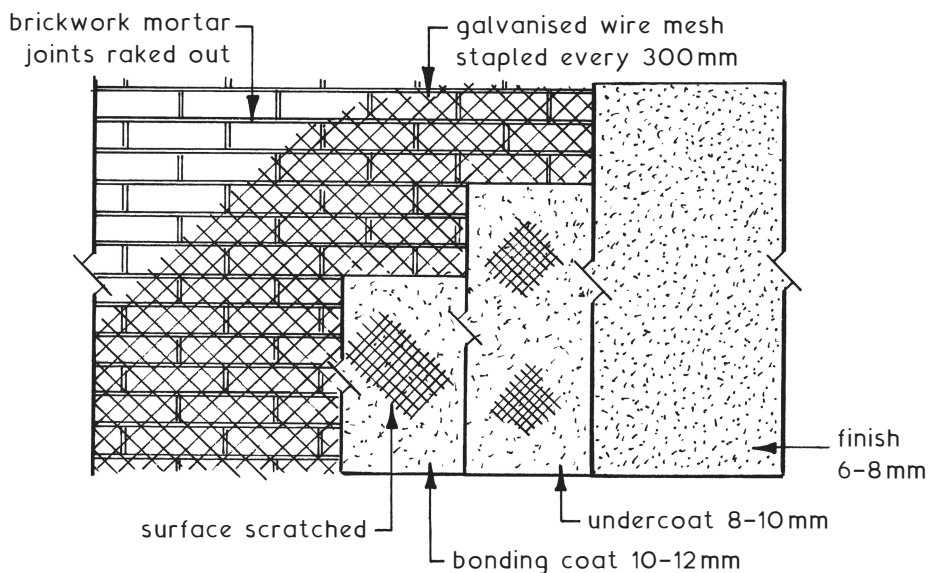
Background ~

Masonry – brick and block-work joints raked out 12 to 15 mm to provide a key for the first bonding coat. Metal mesh can be nailed to the surface as supplementary support and reinforcing.

Wood or similar sheeting – metal lathing, wire mesh or expanded metal of galvanised (zinc coated) or stainless steel secured every 300 mm. A purpose-made lathing is produced for timber-framed walls.

Concrete – and other smooth, dense surfaces can be hacked to provide a key or spatter-dashed. Spatter-dash is a strong mix of cement and sand (1:2) mixed into a slurry, trowelled roughly or thrown on to leave an irregular surface as a key to subsequent applications.

Three coat application to a masonry background



Number of coats (layers) and composition ~ in sheltered locations, one 10 mm layer is adequate for regular backgrounds. Elsewhere, two or possibly three separate applications are required to adequately weatherproof the wall and to prevent the brick or block-work joints from 'grinning' through. Render mixes should become slightly weaker towards the outer layer to allow for greater flexure at the surface, i.e. less opportunity for movement and shrinkage cracking.

Finishes ~ smooth, textured, rough-cast and pebble-dashed.

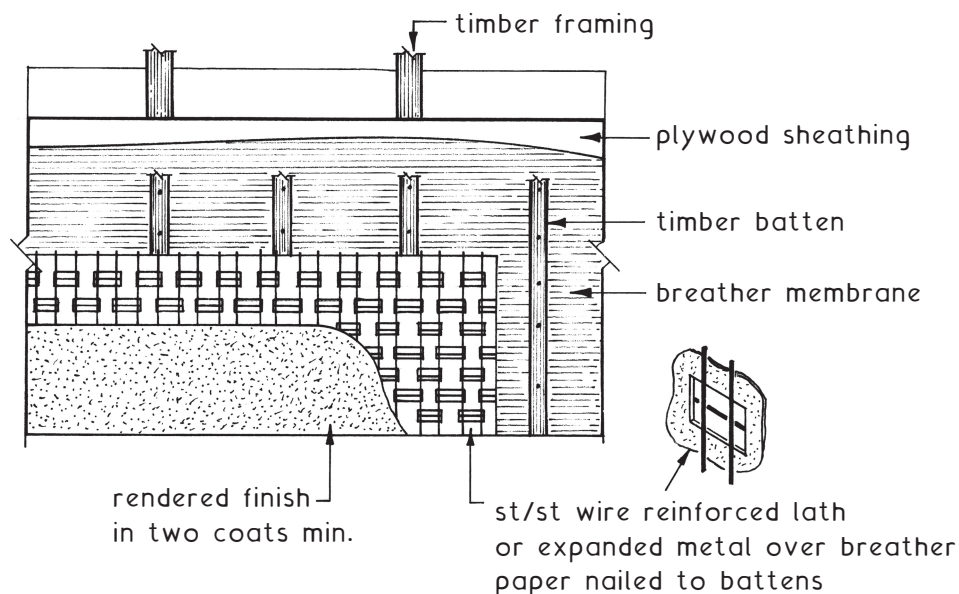
Smooth - fine sand and cement finished with a steel trowel (6 to 8 mm).

Textured - final layer finished with a coarse brush, toothed implement or a fabric roller (10 to 12 mm with 3 mm surface treated).

Rough-cast - irregular finish resulting from throwing the final coat onto the wall (6 to 10 mm).

Pebble or dry dash - small stones thrown onto a strong mortar finishing coat (10 to 12 mm).

Render application to a timber framed background



Ref. BS EN 13914 - 1: Design, preparation and application of external rendering and internal plastering. External rendering.

Traditional renders ~ a site-mixed sand base with a cement and/or lime binder combined with water to produce a workable mix. Mostly used to cover poor quality or inexpensive backgrounds, producing a reasonably waterproof outer surface for decorative treatment. Can also be a design choice.

Polymer modified renders ~ delivered to site in factory-premixed 20 or 25kg bags for the addition of water prior to application. Recipes vary, but in principle manufacturers combine polymeric additives with the base materials of sand and cement. These may include latexes or emulsions, water soluble polymer powders and resins. Polyvinyl acetate (PVA) is the synthetic adhesive generally used. PVA is more commonly known as a wood glue, but in modified renders it functions as an emulsion in water to partially replace cement. Some products also include anti-crack fibres as a reinforcement.

Background ~ existing or newly constructed substrate of render, masonry, concrete, timber or insulation slab (see page 599). Must be sound, reasonably level, free of friable material, algae, lichens and any other surface growths. A fungicidal wash and stabilising solution/primer can be applied to control surface suction.

Application ~ two-coat work for irregular surfaces and refurbishment, 8-10mm base coat reinforced with galvanised or stainless steel mesh secured to the wall. Woven glass-fibre mesh can also be used, generally embedded in the base coat over existing cracks or damaged areas.

Finish ~ 6-8mm over scratched base coat. Surface treatments as described for traditional renders.

Properties ~

- Enhanced weatherproofing.

- Vapour permeable.

- Good resistance to impact.

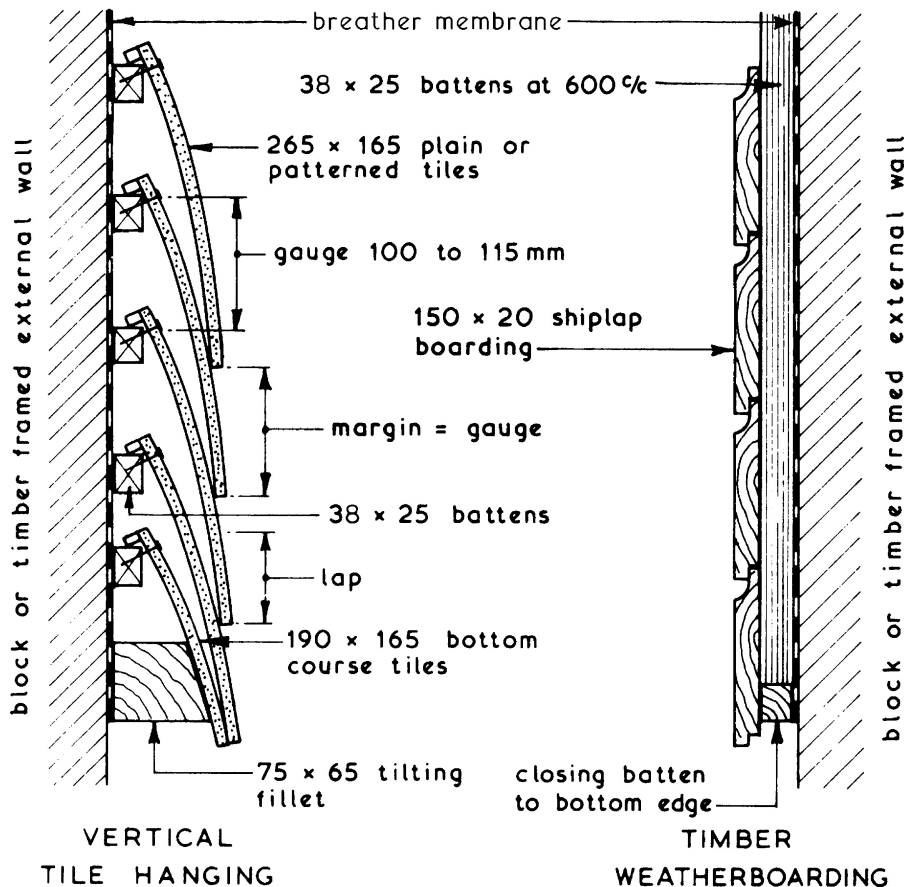
- Inherent adhesion.

- Consistent mix produced in quality controlled factory conditions.

- Suitable for a variety of backgrounds.

Claddings to External Walls ~ external walls of block or timber frame construction can be clad with tiles, timber boards or plastic board sections. The tiles used are plain roofing tiles with either a straight or patterned bottom edge. They are applied to the vertical surface in the same manner as tiles laid on a sloping surface (see pages 521 and 524) except that the gauge can be wider and each tile is twice nailed. External and internal angles can be formed using special tiles or they can be mitred. Timber boards such as matchboarding and shiplap can be fixed vertically to horizontal battens or horizontally to vertical battens. Plastic moulded board claddings can be applied in a similar manner. The battens to which the claddings are fixed should be treated with a preservative against fungi and beetle attack and should be fixed with corrosion resistant nails.

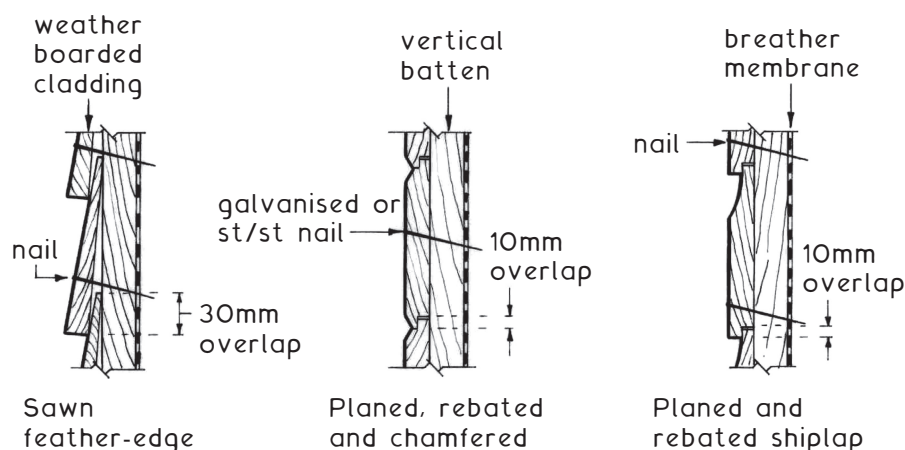
Typical Details ~



Shrinkage ~ timber is subject to natural movement (page 152). With slender cladding sections allowance must be made to accommodate this by drying to an appropriate moisture content before fixing, otherwise gaps will open between adjacent boards.

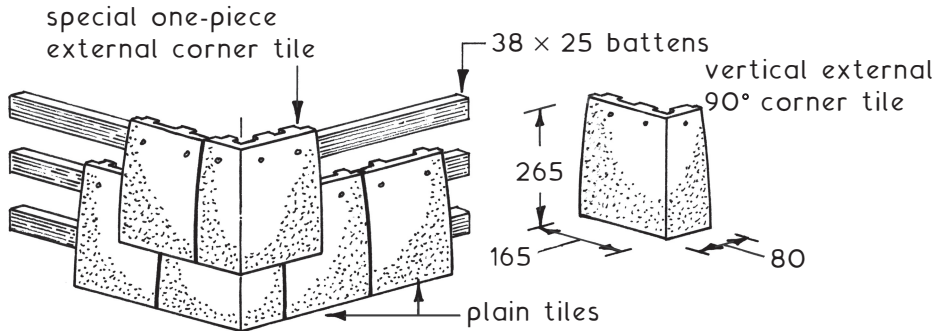
Fixing ~ round head galvanised or stainless steel nails will avoid corrosion and metal staining. Lost head nails should be avoided as these can pull through. Annular ring shank nails provide extra grip. Pneumatic gun fixing to be applied with care to ensure no surface damage from impact. Nail length to penetrate the support battens by at least 22 mm. Two and a half times board thickness is usually specified. Dense timbers such as Siberian larch, Douglas fir and hardwoods should be pre-drilled 2 mm over nail diameter. Double nailing may be required in very exposed situations.

Various Timber Cladding Profiles ~

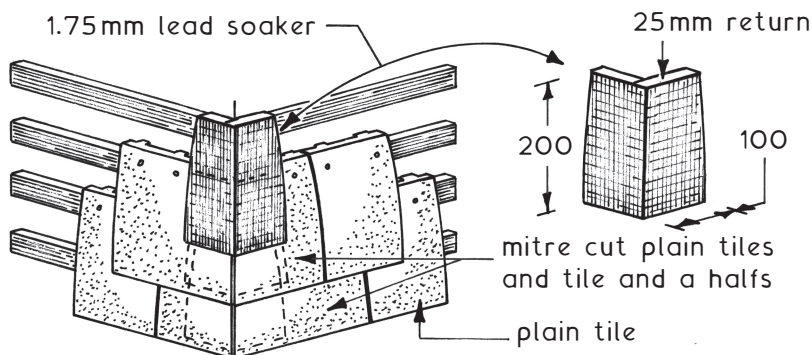


Ref. BS 1186-3: Timber for and workmanship in joinery. Specification for wood trim and its fixing.

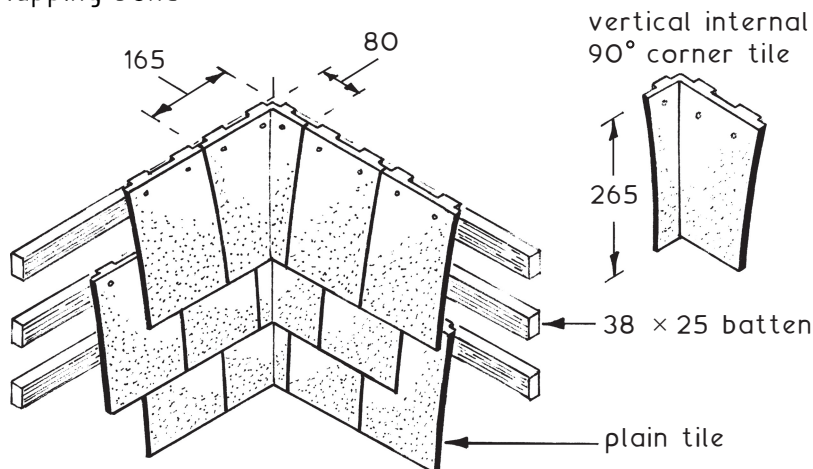
External corner tiles are made to order as special fittings to standard plain tiles. In effect they are tile and a half turned through 90° (other angles can be made) and handed left or right, fixed alternately to suit the overlapping pattern.



An alternative is to accurately mitre cut the meeting sides of tiles. This requires pairs of tile and a half in alternate courses to maintain the bond. Lead soakers (1.75 mm) of at least 225×200 mm are applied to every course to weather the mitred cut edges. The top of each soaker is turned over the tiles and the bottom finished flush or slightly above the lower edge of tiles.

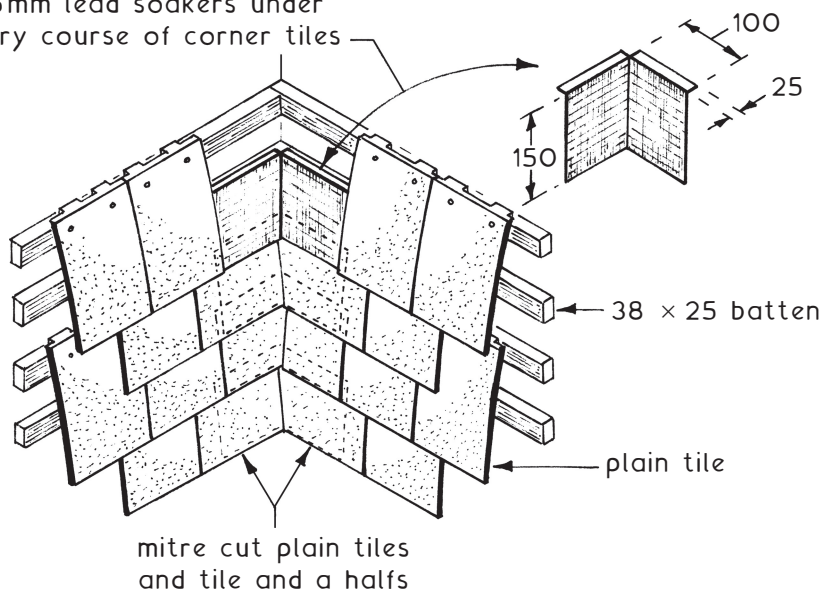


Internal corners are treated similarly to external corners by using special tiles of approximately tile and a half overall dimensions, turned through 90° (or other specified angle) in the opposing direction to external specials. These tiles are left and right handed and fixed alternately to vertical courses to maintain the overlapping bond.

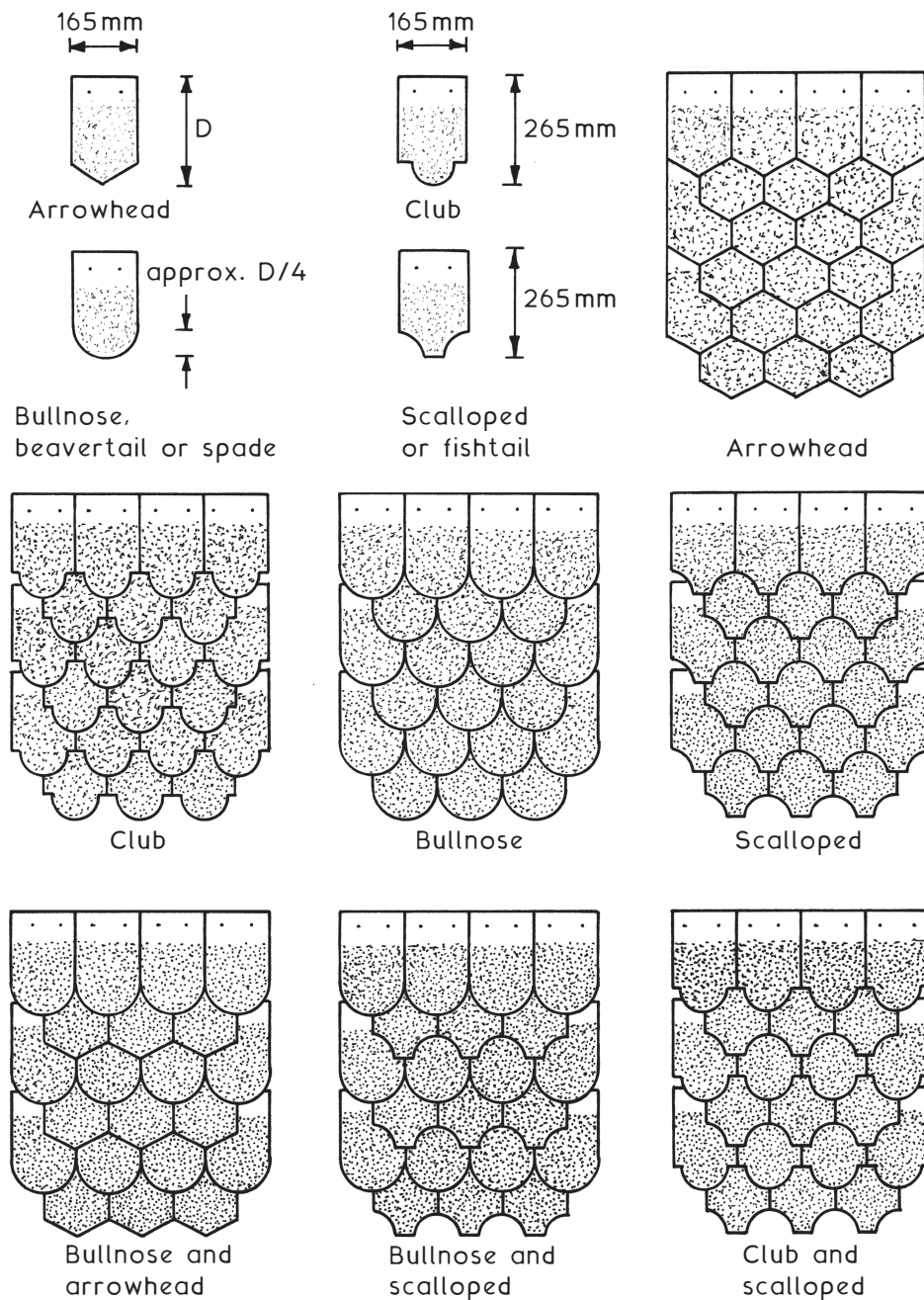


Internal angles can also be formed with mitre cut tiles, with tile and a half tiles in alternate courses. Lead soakers (1.75 mm) of 175×175 mm minimum dimensions are placed under each pair of corner tiles to weather the cut edges. As with external soakers, the lead is discretely hidden by accurate mitre cutting and shaping of tiles.

1.75mm lead soakers under every course of corner tiles



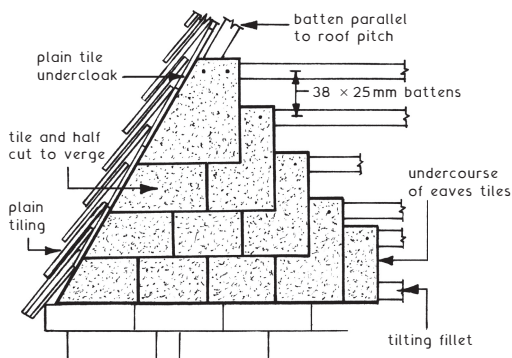
Some standard patterned bottom edge tiles ~



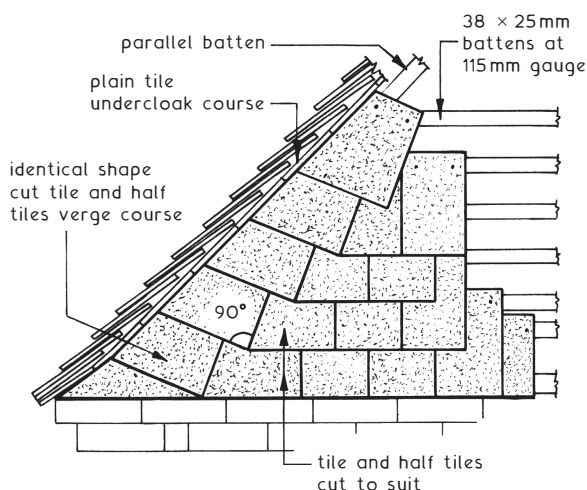
NB. May also be used as a roof-tiling feature, generally in two to three courses at 1.5 to 2.0 m intervals to relieve plain tiling.

Ordinary splay cutting to roof verge ~

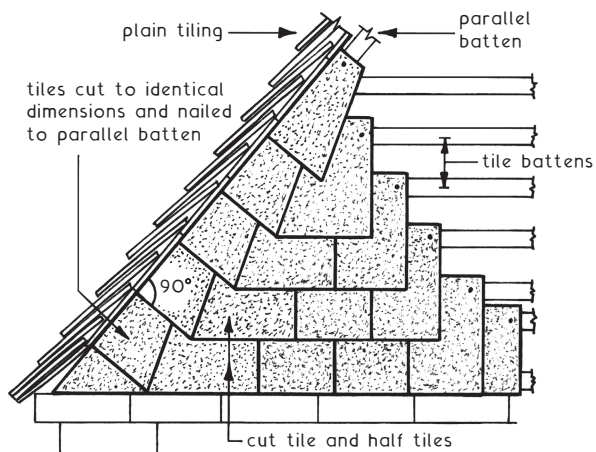
Tile and a half tiles at the end of each course, cut to the undercloak course of roof tiling. A second nail hole can be drilled at the head of each cut tile for secure fixing to the parallel batten.



Winchester cutting to roof verge ~



More attractive than ordinary splay cutting. Two cut tiles are required at the end of each course, varying in size depending on the roof pitch. Tile and half tiles should be used for these to avoid narrow cuts.



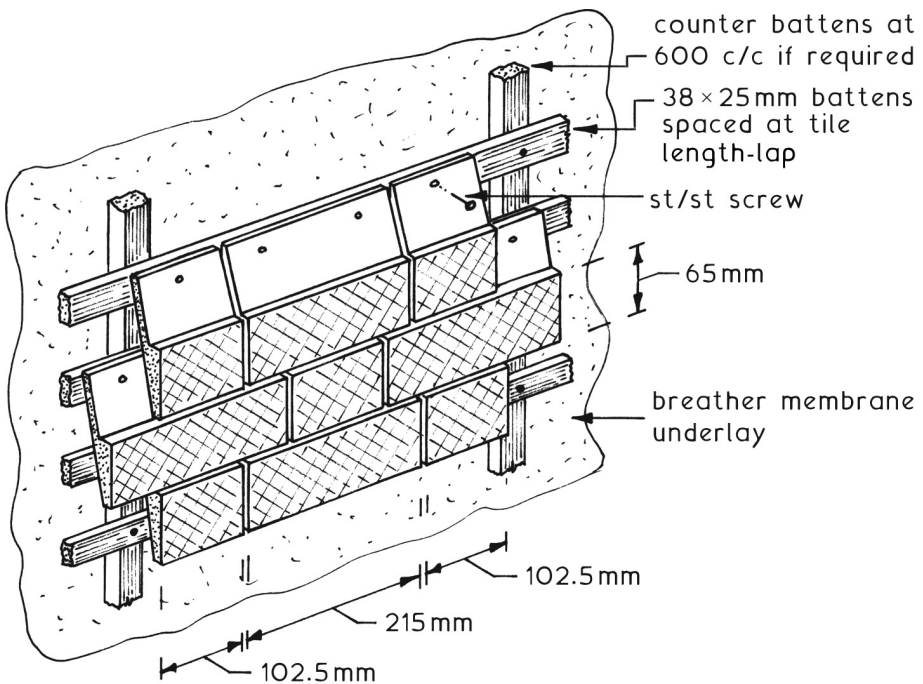
A variation that also has two cut tiles at the end of each course, has the square (90°) edge of the first course of tiles next to the roof tiling undercloak course.

Appearance and concept ~ a type of fake brickwork made up of clay tiles side and head lapped over each other to create the impression of brickwork, but without the expense. Joints/pointing can be in lime mortar or left dry.

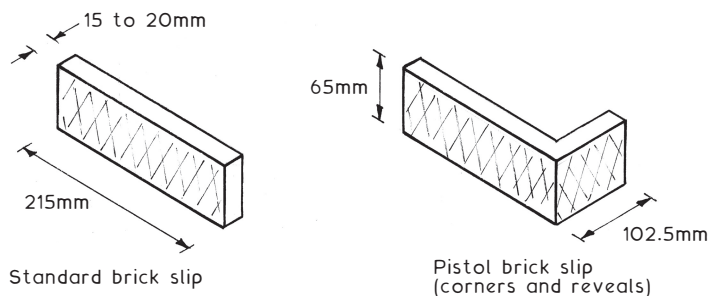
History ~ originated during the eighteenth century, when they were used quite frequently on timber framed buildings notably in Kent and Sussex. Possibly this was to update and improve deteriorated weatherboarding or to avoid the brick tax of 1784. This tax was repealed in 1835.

Application ~ restoration to mathematically tiled older structures and as a lightweight cladding to modern timber-framed construction where the appearance of a brickwork façade is required.

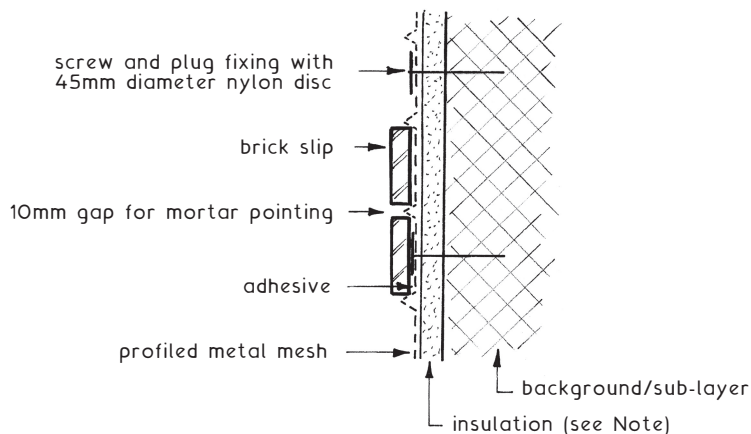
Application shown in a Flemish bond ~



Brick slips ~ thin bricks resembling tiles of 15 to 20mm thickness with face dimensions the same as standard bricks (see page 404) and a finish or texture to match. Other terminology includes brick tiles, brick cladding units, slip bricks and briquettes.



Application ~ used to maintain continuity of appearance where brickwork infill panels are supported by a reinforced concrete frame, as shown on page 739. Originally slips were cut from whole bricks, a time consuming and wasteful process no longer necessary, as most brick manufacturers make slips to complement their stock items. Other uses are as a cost effective dressing to inexpensive or repaired backgrounds and for use where it is too difficult to build with standard bricks.



Support system ~ profiled galvanised or stainless steel mesh secured with screws and nylon discs 300mm horizontally and 150mm vertically apart into a sound background. Slips are stuck to the mesh with a structural adhesive of epoxy mortar, a two-part mix of resin and hardener combined with a sand filler.

Note: Extruded polystyrene closed cell foam insulating panel 25 to 50mm in thickness, adhesive secured to the background.